

Week 3 — Input & Text → Mad Libs

Name: ____ Date: ____

`input()` asks and **waits**. Numbers come back math-ready. Words need the `'s'` flag — otherwise MATLAB reads them as **code**.

Today's words

Word	What it means
<code>input()</code>	asks a question, then waits (forever, if it has to) for an answer
the 's' flag	<code>input('...', 's')</code> — expect WORDS, not code
<code>fprintf</code>	the Mad Libs machine — a sentence with slots, filled in order
slot	<code>%s</code> = word slot, <code>%d</code> = number slot
<code>\n</code>	"press Enter" at the end of an <code>fprintf</code> sentence
<code>num2str()</code>	turns a number into text so <code>[]</code> can glue it

1 • Be the computer 🤖

The user types **walrus** when asked. Write what each output line shows. (Careful — one of these commands has no slots to fill!)

```
animal = input('Animal? ', 's');
fprintf('A %s ate my homework!\n', animal)
disp('A %s ate my homework!')
```

Line 1: ____

Line 2: ____

2 • Adding or gluing? 🧩

Write what each line displays:

```
disp(3 + 3)           % → ____
disp(['3', '3'])      % → ____
disp(['ha', 'ha'])    % → ____
disp(['1', '2'])      % → ____ (careful — glue, not math!)
```

3 • Spot the bug 🐛

This program should greet Ada — but it CRASHES on line 1 with `Unrecognized function or variable 'Ada'.` Circle the problem, then write the fixed line 1 on the blank:

```
name = input('Name? ');           % user types: Ada
fprintf('Hi, %s!\n', name)
```

Fixed line 1: `name = input( _____ )`

🧠 Brain teaser (optional — take it home)

Design your own Mad Libs! Write a 3–5 sentence story with **at least four blanks** (noun, verb, adjective, number — your choice). For each blank, write the exact question the computer should ask, like:

```
food = input('Give me a food: ', 's')
```

Then write your story as `fprintf` lines — `%s` where a word goes, `%d` where a number goes, `\n` at the end of each sentence. Bonus points if your number blank does MATH inside the story, like `fprintf('...%d...', number * 100)` — remember: the number question is the one WITHOUT the `'s'`.

Bring it next week — the best one gets typed in and **run on the smartboard**, starring your words. Fame awaits.